THE PRESIDENT'S OFFICE – RALG

TAHOSSA ROMBO DISTRICT EXAMINATION SYNDICATE

FORM FOUR MOCK EXAMINATION

032/2B **CHEMISTRY 2B (ACTUAL PRACTICALS)**

TIME: 3:00 Hours **MARKING SCHEME** **August, 2024**

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1. $Volume\ of\ burette\ used = 50cm^3$

$Volume\ of\ pipette\ used = 20cm^3$

$Table\ of\ results$

Burette readings(cm^3)	Pilot	1	2	3
Final burette reading	16.20	32.30	16.00	15.90
Initial burette reading	0.00	16.20	0.00	0.00
Volume of acid used	16.20	16.10	16.00	15.90

*1mark**@=04 total marks***

To calculate average volume of acid used

$V_A = \left[\frac{V_1 + V_2 + V_3}{3}\right] cm^3$ ……0.5mark

$$V_A = \left[\frac{16.10 + 16.00 + 15.90}{3}\right] cm^3$$

$$V_A = \left[\frac{48}{3}\right] cm^3$$

$$V_A = 16cm^3$$

∴ The average volume of acid used is $16cm^3$……..0.5mark

Note: *If the pipette used is* $25cm^3$ *the average volume of acid used will be* $20cm^3$

a) $20cm^3\ of\ solution\ NN\ required\ 16cm^3$ of solution MM for complete neutralizati
*1mark**@=02total marks***

b) The solutions will be contaminated and also may lead the change in concentration due to the presence of water droplets……02marks

c) Solution MM is dilute hydrochloric acid (HCl)
Solution NN is sodium carbonate solution (Na_2CO_3)

$$Balanced\ chemical\ equation: 2HCl_{(aq)} + Na_2CO_{3(aq)} \rightarrow 2NaCl_{(aq)} + H_2O_{(l)} + CO_{2(g)} 03marks$$

d)

i. First stage: To calculate number of moles of hydrogen chloride
Data

$$Volume(V) = 0.56dm^3$$

$$GMV\ at\ S.T.P = 22.4dm^3$$

$$Required: Number\ of\ moles\ (n)$$

$From\ the\ formula: Number\ of\ moles\ (n) = \frac{Volume(V)}{GMV\ at\ S.T.P}$……..01mark

$$n = \frac{0.56dm^3}{22.4dm^3}$$
$$n = 0.025mol$$

$\therefore$ *The number of moles of hydrogen chloride is* $0.025mol$……..01mark

Second stage: *To calculate molarity of solution MM*

Data

$$Number\ of\ moles\ (n) = 0.025mol$$
$$Volume\ of\ solution(V) = 250cm^3$$
$$\approx 0.25dm^3 \ldots\ldots\ldots 0.5mark$$
$$Required: Molarity(M)$$

$From\ the\ formula: Molarity(M) = \frac{Number\ of\ moles(n)}{Volume\ of\ a\ solution\ (V)}$……..01mark

$$M = \frac{0.025mol}{0.25dm^3}$$
$$M = 0.1mol/dm^3$$

$\therefore$ *The molarity of solution MM is* $0.1mol/dm^3$…….01mark

ii. Solution

Data

$$M_A = 0.1mol/dm^3$$
$$V_A = 16cm^3$$
$$V_B = 20cm^3$$
$$2HCl_{(aq)} + Na_2CO_{3(aq)} \rightarrow 2NaCl_{(aq)} + H_2O_{(l)} + CO_{2(g)}$$
$$n_A = 2mol$$

$n_B = 1mol$ ……..0.5mark

$$Required: M_B$$

$\frac{M_AV_A}{M_BV_B} = \frac{n_A}{n_B}$, *To make "*$M_B$*" the subject* ……01mark

$$M_B = \frac{M_AV_An_B}{V_Bn_A}$$
$$M_B = \frac{0.1\ X\ 16X1}{20X2}$$
$$M_B = \frac{1.6}{40}$$
$$M_B = 0.04mol/dm^3$$

$\therefore$ *The molar concentration of solution NN is* $0.04mol/dm^3$……01mark

iii. Solution

Data

Mass (m)=2.86g

Volume of a solution(V)=$0.25dm^3$

Required: Mass concentration (C)

From the formula: $Mass\ concentration\ (C) = \frac{Mass\ (m)}{Volume\ of\ a\ solution\ (V)}$……01mark

$$C = \frac{2.86}{0.25}$$
$$C = 11.44g/dm^3$$

$\therefore$ *The mass concentration of solution NN is* $11.44g/dm^3$……01mark

iv. First stage: To calculate molar mass of solution NN

Data

$$Mass\ concentration\ (C) = 11.44g/dm^3$$
$$Molar\ concentration(M) = 0.04mol/dm^3$$
$$Required: Molar\ mass\ (mr)$$

$From\ the\ formula: Molar\ concentration\ (M) = \frac{Mass\ concentration\ (C)}{Molar\ mass(mr)}$……01mark

$$Molar\ mass\ (mr) = \frac{Mass\ concentration\ (C)}{Molar\ concentration(M)}$$

$$Molar\ mass\ (mr) = \frac{11.44}{0.04}$$

$$Molar\ mass = 286g/mol$$

$\therefore Molar\ mass\ of\ solution\ NN\ is\ 286g/mol$……01mark

$Second\ stage: To\ calculate\ number\ of\ molecules\ of\ crystallization$

$Na_2CO_3.WH_2O = mr$……01mark

(23X2) + 12 + (16X3) + W(18)=286

46+12+48+18W=286

106+18W=286

18W=286-106

$$\frac{18W}{18} = \frac{180}{18}$$

W=10

$\therefore Number\ of\ molecules\ of\ crystallization\ in\ hydrated\ sodium\ carbonate\ is$ 10

01mark

2.

S/N	Observation	Inference
(a)	(i)Colour White colour observed (ii) Texture Crystalline form (iii)Deliquescence Absorbed water from the atmosphere	 $NH_4^+, Na^+, Ca^{2+}, Zn^{2+}, Pb^{2+}$may be present NO_3^-, SO_4^{2-}, Cl^- may be present $NO_3^-, Cl^-, SO_4^{2-},$ may be present
(b)	Cracking sound with brown gas observed	$NO_3^-\ of\ Pb^{2+}$may be present
(c)	No gas evolved	$SO_4^{2-}, NO_3^-Cl^-$ may be present
(d)	(i) White precipitate was formed, soluble in excess.	Zn^{2+}, Pb^{2+}may be present
	(ii) White precipitate was formed, insoluble in excess	Pb^{2+} may be present
	(iii) Yellow precipitate which disappeared on warming but reappeared on cooling.	Pb^{2+} confirmed
	(iv) Brown ring was formed at the junction of the liquids.	NO_3^- confirmed

*1mark***@=18 total marks**

i. Cation present in sample Y is *while* anion is NO_3^- *1mark***@=02 total marks**

ii. Lead (II) nitrate…….01mark

iii. Experiment (b)

$2Pb(NO_3)_{2(s)} \xrightarrow{\Delta} 2PbO_{(s)} + 4NO_{2(g)} + O_{2(g)}$……..02marks

$Experiment\ (d) - iii$

$Pb(NO_3)_{2(aq)} + 2KI_{(aq)} \rightarrow PbI_{2(s)} + 2KNO_{3(aq)}$……..02marks